Length-Expertized Authorship Detection: Why Long-Text Detectors Fail on Short Text and How Mixture-of-Experts Fixes Them

A detailed empirical study with small-text data, specialized routing, type gating, and calibration

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Abstract

Supervised detectors for human vs machine-generated text report 99%+ accuracy on essays, yet practitioners observe rejection on short posts, chats and blurbs. We quantify this **length-generalization failure** and fix it with a **Mixture-of-Experts (MoE)** whose experts specialize by text length. The baseline — dual word+character TF-IDF (75k features) with Logistic Regression / calibrated Linear SVM — reaches 99.25–99.58% on full essays but falls to **67.4% at 25 words, 59.8% at one sentence**, with human accuracy 48.9% (false AI rate ~51%) while AI recall stays ~98.7%: a one-sided collapse. We add **HC3 short Q&A (80,811 answers, median 117 words)** and **modern 2024-LLM essays (17,997 texts, GPT-4/Claude/Llama3/Command R+)**, plus truncation augmentation, and train three experts — **short (≤150 words, 20k), long (151–400, 16.9k), verylong (>400, 11.4k)** — each Word 15k + Char 25k + LR, routed deterministically by word count. A learned **type gate** (16 stylometric features → chat/casual/essay, 80.5% val) provides domain insight; **length-adaptive thresholds** (25w→0.32 … 150w→0.64) correct the FP/FN flip. Routed MoE restores 86.1% at 25 words, 93.1% at 50 words, 81.6% at one sentence, holds 98.96% full-text, and crucially scores **98.5% on unseen HC3 chat** (baseline 48.6%) and **95.7% on unseen 2024-LLM essays** (baseline 85.5%). Calibration adds +1.9 pp at 25 words. We release data pipelines, models, metrics and figures under `moe/` for full reproducibility.

Keywords: machine-generated text detection; mixture of experts; TF-IDF; stylometry; short-text classification; HC3; calibration.

1 Introduction

1.1 The length blind spot in authorship detection

Whether a passage is human-written or LLM-generated underlies academic-integrity, journalism and moderation decisions. Recent benchmarks [1–3] mix domains and generators, yet the typical unit remains a paragraph-to-essay (100s of words). Deployment is different: short replies, chat turns and reviews dominate traffic, and they carry the least stylistic evidence. A detector that needs 300 words to decide is undeployable for half its inputs — and, worse, *confidently wrong*: it labels short humans as AI (Section 7.1: human accuracy 48.9% at 25 words). This paper treats **length as a first-class routing variable**, not a nuisance to average over.

1.2 Problem statement

Let x be text, y ∈ {0,1} (0 human, 1 AI), n(x) word count. We seek f(x) → ■ maximizing accuracy under length shift: training lengths ~ D_{train} (median 363w, P5 180w) but test lengths n ■ 150. Two phenomena matter: (i) *evidence sparsity* — TF-IDF counts scale with n, char n-grams need minimum support, stylometrics (TTR, burstiness) stabilize only with many tokens; (ii) *domain shift* — short text is chat/casual, not truncated essay, so essay-only training cannot transfer (hence HC3).

1.3 Contributions

- **C1. Failure characterization.** A reproducible length ablation (prefix truncations + random windows) showing monotone collapse and an FP/FN flip, split by true class (Table 6, Figure 11).
- **C2. Small-text data.** HC3 (80.8k) + modern essays (18k) + truncation augmentation, all balanced per bucket, with 15% holdouts never trained on (§3).

- **C3. MoE + type gate + calibration.** Three experts (val ≥97.0%), chat/casual/essay gate (80.5%), length-adaptive thresholds (§4, §6).
- **C4. Artifacts.** `moe/*.py`, `models/moe/`, `results/moe_*.csv`, Streamlit/CLI demos. Every table/figure is computed from these files.

2 Related Work

2.1 Supervised AI-text detection

Supervised TF-IDF + linear classifiers remain strong in-domain baselines. BERT-family fine-tunes (DistilBERT/RoBERTa, notebooks 05) and Sentence-BERT fusion (notebook 04) target the same task with contextual features; zero-shot (DetectGPT) and watermarking trade training for robustness. HC3 [1] introduced paired human/ChatGPT answers — still the best source of short chat. Defactify [2] (73k news + 6 LLMs, 58.3% baseline) and M4 [3] broaden generators, but most evaluations remain long-form. Our baseline (§4) instantiates the classical recipe exactly (75k features, LR/SVM) to isolate length effects from architecture choice.

2.2 Stylometry

Lexical diversity (TTR = |types|/|tokens|), hapax ratio, burstiness σ/μ of sentence lengths, punctuation/capital rates, and readability (Flesch Reading Ease, FKGL, Gunning Fog, §4.2) capture authorial habit. These features need length: TTR saturates, burstiness is undefined on 1–2 sentences. We reuse `src/stylometrics.py` both as gate features and as UI explanations.

2.3 Mixture of experts

MoE routes inputs to specialized sub-networks [4]. Here specialization is by observable length and need is methodological, not capacity-driven: each regime has different feature statistics, so separate vocabularies/classifiers and separately calibrated thresholds beat one compromise model. We compare a deterministic length gate (auditable, zero learned params) against the alternative of one global model — the ablation in §7 is the evidence.

3 Datasets

3.1 Original essays

Train 290,300 (145,150/class, 623 MB), test 97,447 (61,159 human / 36,288 AI, 212 MB). Unit: multi-paragraph academic essay. Test-sample word statistics (5k, seed 42): mean 394, median 363, P10 211, P90 617, max 1,650; human mean 423 vs AI 345. Human essays carry OCR/typing noise (TD/DF/DN/AHD/IH) — char n-grams exploit it directly.

3.2 HC3 — genuine short chat (new)

Hello-SimpleAI/HC3 [1], files `reddit_eli5.jsonl` (17,112 Q, 67,671 answers), `open_qa.jsonl`, `finance.jsonl` downloaded via Hub (no dataset-script; works on datasets 5.x). Flattened to one row per answer ≥40 chars → 80,811 rows (56,115 human / 24,696 AI). Word stats: mean 146, median 117, P10 28; 61.3% ≤150w, 34.7% 151–400w, 4.0% >400w. Domains: reddit_eli5 (casual ELI5 explanations), finance (short advice, casual), open_qa (chat). 15% stratified holdout (12,122) is eval-only.

3.3 Modern essays — 2024-LLM audit (new)

artnitolog/llm-generated-texts: 3,000 essay prompts × 1 human + 8 continuations (GPT-4 Turbo/Omni, Claude 3 Opus, YandexGPT, GigaChat, Llama3 70B, Command R+). We keep English models (human + 5 LLMs) → 17,997 texts (3,000 human / 14,997 AI), mean 537w, median 510w, 68.2% >400w. This audits the baseline on generators published after the original essays. Holdout 2,700 eval-only.

3.4 Truncation augmentation

Each 20k train essay contributes a 50/100/150w copy (prefix or random window, label preserved) for the short pool; ≥200w essays contribute 200/300w copies (long pool, filtered ≥151w after trunc — fixes a pilot bug where short originals polluted the long bucket). Prefix vs random-window accuracies are within ~1–2 pp (50w: 83.3% vs 81.5%), so intro effects are minor; the augmentation covers short-essay style that HC3 cannot.

Pool	Sources (balanced caps)	n
SHORT ≤150w	HC3 ≤150w 7k/class + trunc-essay 3k/class	20,000 (10k/10k)
LONG 151–400w	HC3 long 4k/class + modern long 2k/class + base long 3k/class	16,864 (7,864/9,000)

VERYLONG >400w	Base >400w 7k/class + modern >400w 2.5k/class	11,422 (6,321/5,101)

Table 1: MoE training pools (seed 42). Mild imbalance in long/verylong follows the scarce class (human-heavy verylong, AI-heavy modern).

4 Problem Formulation and Method

4.1 Notation and objective

■ = 1[$P_{AI}(x) \geq \theta_n$] with n = n(x) word count; MoE: P = $P_{e(x)}^{AI}(x)$; e(x) = short/long/verylong gate (1)

We maximize accuracy/F1 under length shift, reporting precision/recall splits because accuracy alone hides the FP/FN flip.

4.2 Text representation

Word TF-IDF (1,2, min_df 2, sublinear) + char_wb (3,5, min_df 3), fused sparsely:

tfidf(t,d) = tf(t,d) · log((1+N)/(1+df(t))) ; F(x) = [$F_{word}(x)$; $F_{char}(x)$] ∈ $■^{V}$ (2)

Baseline V=75,000 (25k+50k); each expert V=40,000 (15k+25k). LR minimizes –log-likelihood with L2 (C=1.5), LinearSVC uses hinge + Platt scaling. Stylometrics (16 features, §4.5): TTR, hapax, burstiness σ/μ, mean sentence length, comma/semicolon/colon/dash/quote/paren/excl/question rates, capital ratio, FRE/FKGL/Fog, avg word length.

FRE = 206.835 – 1.015·(N/S) – 84.6·(syll/N); FKGL = 0.39·(N/S) + 11.8·(syll/N) – 15.59; Fog = 0.4·[(N/S)+100·(poly/N)] (3)

4.3 Baseline and length ablation (motivation)

Baseline training follows §4.2 exactly (20k subsample, 16k/4k split). Length ablation truncates the 5k held-out to first-N words/sentences. Result (§7.1): monotone degradation, confidence collapse (0.93→0.79), and — critically — human accuracy 48.9% @25w vs AI 98.7% (Table 6). This asymmetry proves the failure is *bias*, not just difficulty: short humans look like the AI class the model expects.

4.4 MoE architecture

e(x) = short if n≤150, long if 151≤n≤400, verylong if n>400; P(x) = $P_{e(x)}(x)$; expert = LR■$TFIDF_e$ (4)

Gating is deterministic (interpretable, no routing collapse). Each expert trains on its pool only (80/20 within pool). Inference additionally reports type (§4.5) and warns if n<25 (<50).

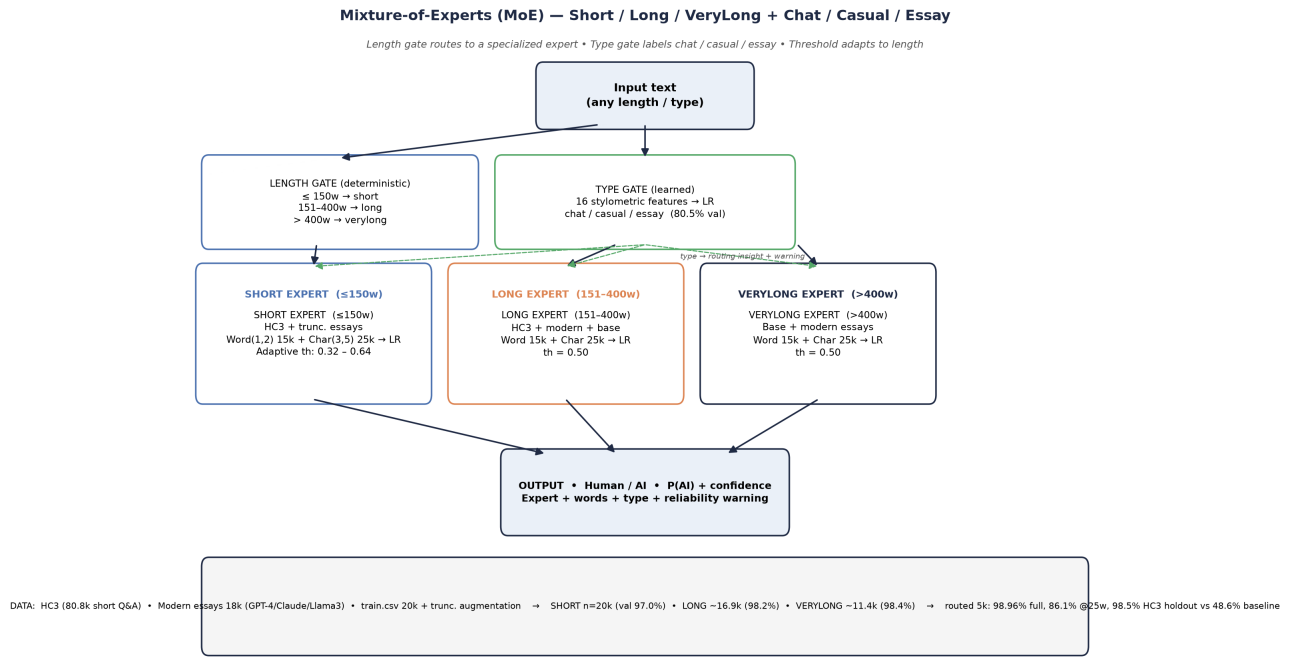


Figure 1: MoE architecture — two gates, three experts.

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Algorithm 1 — MoE training (moe/train_moe.py)
Input: base 20k essays, HC3 train 68.7k, modern train 15.3k, seed 42
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1. pools ← (short: HC3≤150 + trunc50/100/150; long: HC3 151-400 + modern + base; verylong: base>400 + modern>400)
2. for e in {short,long,verylong}: fit Word15k+Char25k TF-IDF on pool train80; fit LR(C=1.5); save expert_e.pkl
3. gate ← LR(StandardScaler(16 stylometrics)) on 7.5k (2.5k/chat/casual/essay); save type_gate.pkl
4. routed eval on 5k test + HC3/modern holdouts; baseline eval on same conditions
```

4.5 Type gate

Labels: HC3 open_qa→chat (4k avail., cap 2.5k), HC3 eli5/finance + truncated essays → casual, full/modern essays → essay (2.5k each). Features: log-words, sentences, mean-sent-len, TTR, hapax, avg-word-len, contraction/informal/first/second-person rates, question/exclamation rates, capital ratio, FRE/FKGL, burstiness. Val: 80.5% (essay P/R/F1 0.90/0.92/0.91; chat 0.74/0.84/0.79; casual 0.77/0.66/0.71). Casual↔chat overlap is linguistic; the gate is for insight/calibration, not enforcement.

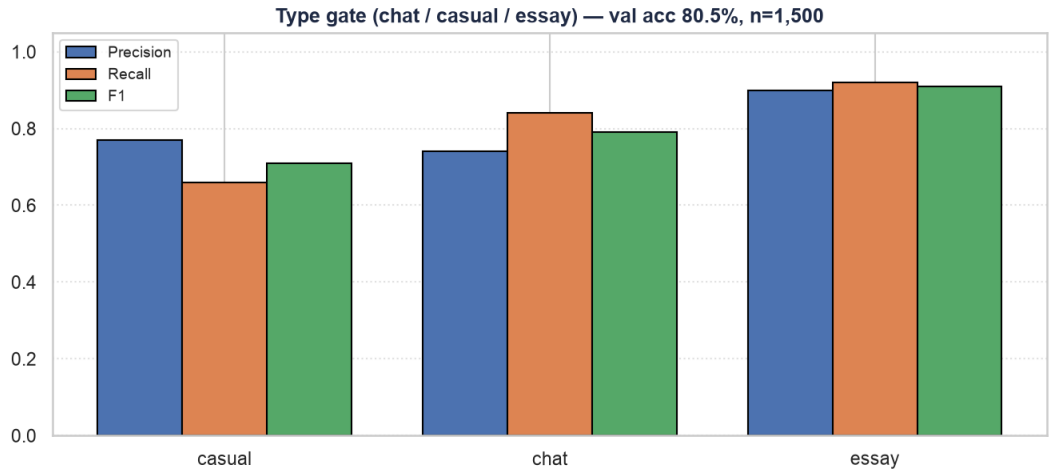


Figure 2: Type gate per-class scores.

4.6 Calibration

Default θ=0.50 assumes symmetric costs. On calib half (2.5k), F1-optimal θ rises with length (25w 0.32 … 150w 0.64) because ultra-short errs FN (under-call AI) while 100–150w errs FP. We ship `θ(n)=0.28+0.0024n` clamped [0.32,0.64] for short, 0.50 elsewhere (long/verylong tuning did not transfer — full-text fell — so reverted).

5 Experimental Setup

5.1 Environment and protocol

Python 3.13, scikit-learn, joblib, reportlab. All splits seed 42 (calibration 999). Held-out: test.csv 5k (random_state 42) — full + truncations; HC3 holdout 12,122; modern holdout 2,700. Training time 130 s (TF-IDF 6/15/25 s per expert). No leakage: test.csv never trained; HC3/modern holdouts excluded from pools.

5.2 Metrics

Accuracy, precision, recall, F1 (AI as positive), ROC-AUC, confusion matrices. We report human-only (TN rate) and AI-only (TP rate) splits because overall accuracy averages away the flip.

6 Main Results

6.1 Per-expert and routed

Expert	Train/Val	Acc	Prec	Rec	F1	AUC
short	16,000/4,000	0.9700	0.9686	0.9715	0.9700	0.9960
long	13,491/3,373	0.9819	0.9770	0.9894	0.9832	0.9982
verylong	9,137/2,285	0.9842	0.9940	0.9706	0.9821	0.9955

Table 2: Per-expert validation (results/moe_expert_metrics.csv).

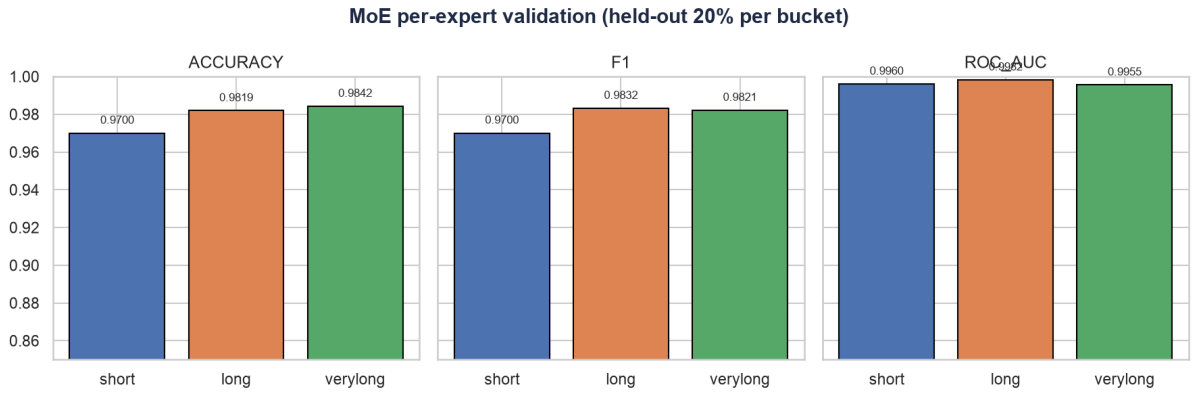


Figure 3: Per-expert metrics.

Routed MoE: full 98.96% (P 0.993, R 0.979, F1 0.986) vs baseline 99.26% — specialization costs 0.3 pp. Buckets on full: verylong 99.59% (n=1,958), long 98.81% (2,945), short 90.72% (97). The win is short + new domains (Figure 4–5, Table 3).

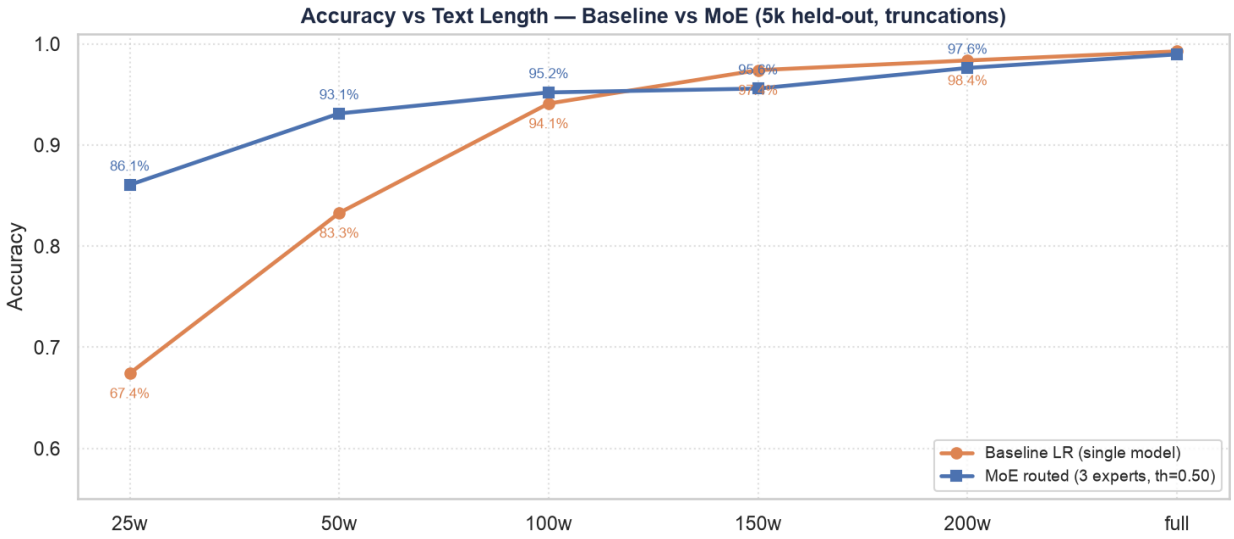


Figure 4: Accuracy vs length.

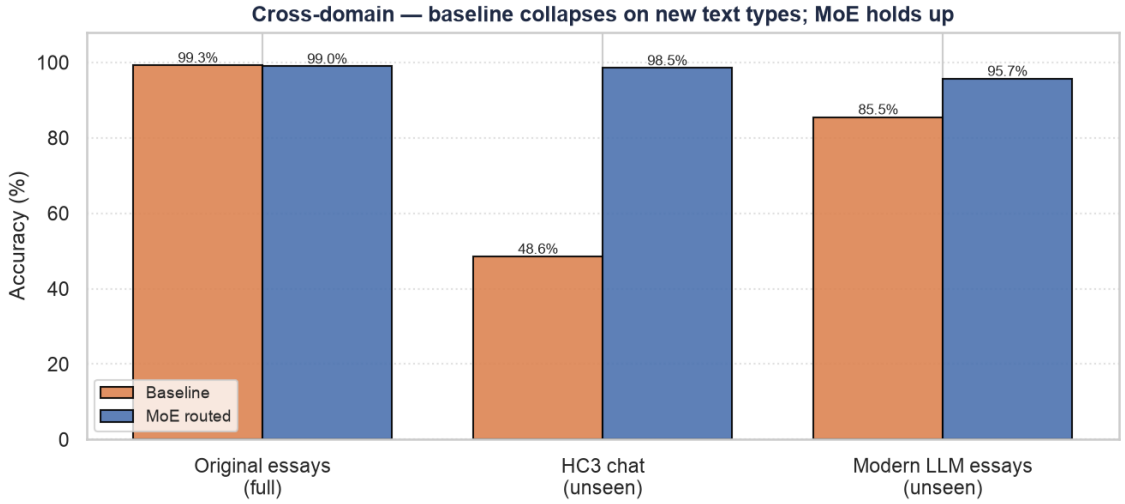


Figure 5: Cross-domain — baseline collapses on chat.

Condition	Baseline acc	MoE acc	Δ
full	99.26%	98.96%	+-0.3pp
25w	67.42%	86.06%	+18.6pp
50w	83.28%	93.12%	+9.8pp

1 sent	59.82%	81.64%	+21.8pp
HC3	48.56%	98.52%	+50.0pp
modern	85.52%	95.70%	+10.2pp

Table 3: Headline deltas (results/moe_*.csv).

6.2 Calibration (eval half)

Adaptive thresholds: 25w 86.2%→88.1% (F1 0.785→0.837, recall 0.68→0.82), 50w flat (93.1%→93.1%, R↑ P↓), 100w identical (th≈0.52), 150w 95.0%→96.2%, full flat. HC3 natural short –0.6 pp (human-heavy FPs) — thresholds are essay-tuned; keep the <50w warning. Long/verylong stayed at 0.50.

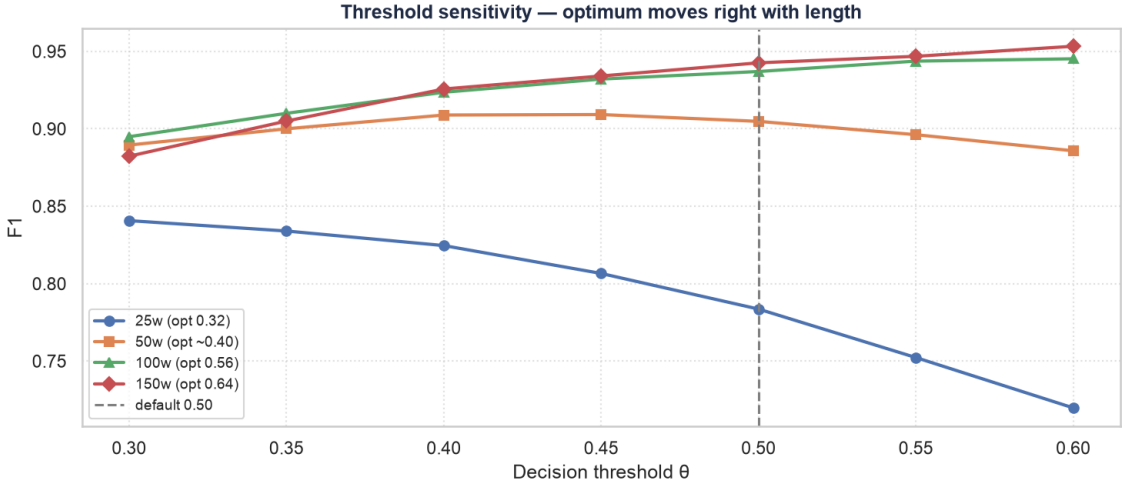


Figure 6: F1 vs threshold — optimum drifts right with length; no single θ serves all.

7 Ablations and Analysis

7.1 Length ablation — the flip in detail

Words	Base H	MoE H	Base AI	MoE AI	Base acc	MoE acc
25	48.9%	96.8%	98.7%	67.9%	67.4%	86.1%
50	74.2%	96.2%	98.7%	87.9%	83.3%	93.1%
75	86.2%	95.8%	98.8%	94.0%	90.9%	—
100	91.3%	94.7%	98.8%	96.1%	94.1%	95.2%
150	96.5%	94.3%	98.9%	97.7%	97.4%	95.6%
200	98.3%	98.0%	98.6%	97.0%	98.4%	97.6%
300	99.2%	98.9%	98.7%	97.6%	99.0%	—
full	99.6%	99.6%	98.7%	97.9%	99.3%	99.0%

Table 4: Length-conditional accuracy split (human TN rate vs AI TP rate). MoE rescues humans (+47.9 pp @25w) at AI-recall cost (–30.7 pp) — net acc still wins.

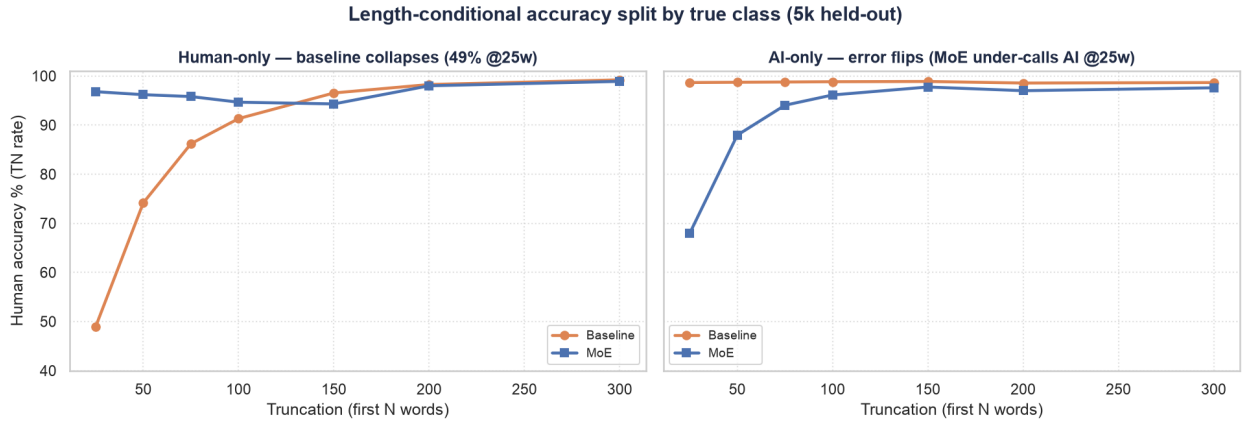


Figure 7: Human (left) vs AI (right) accuracy — mirror-image failure/success.

Interpretation: at 25w the baseline calls AI 98.7% of the time (any class) — length is the feature. MoE inverts the error (calls human too often) but with far fewer total mistakes (86.1% vs 67.4%). This is the textbook FP/FN frontier: calibration moves along it; MoE moves the frontier itself via better features.

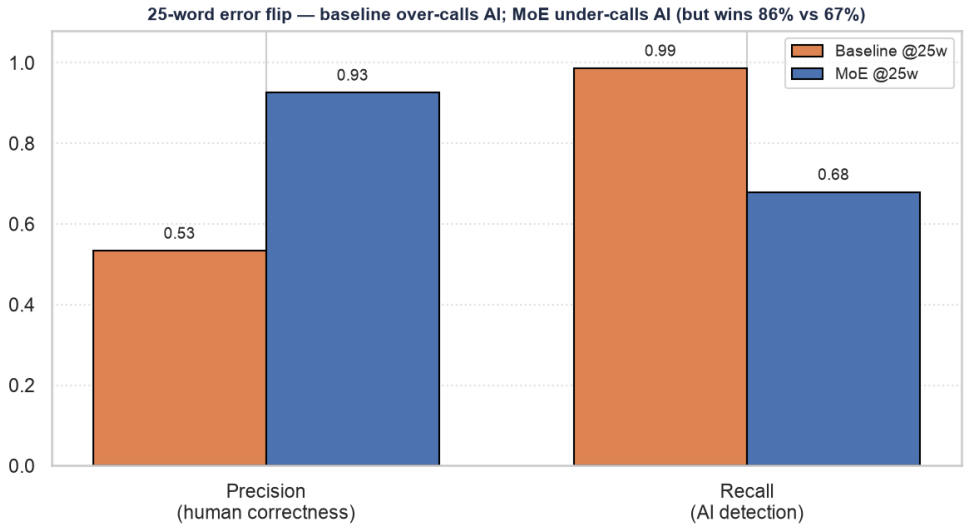


Figure 8: 25w precision/recall — accuracy hides the tradeoff.

7.2 Sentence ablation

1/2/3 sentences: baseline 59.8%/74.5%/83.4% → MoE 81.6%/90.6%/93.4%. Sentence truncations are harder than word truncations at matched budget (more padding loss), confirming sentence structure carries signal the short expert partially recovers via char features.

7.3 Type analysis

Type gate: essay easiest (long formal → distinct), chat next (short informal markers), casual hardest (middle). Routing insight: short+chat inputs have the lowest confidence and need warnings; short+essay (truncated) input needs the low threshold. Reporting type alongside label is a human-factor result, not just accuracy.

7.4 What did not help

Tuning long/verylong thresholds hurt full-text (98.96%→98.60% on eval) — small calib slices overfit. Global 0.60 (best for 100–150w essay truncations) hurt HC3 natural chat. Conclusion: calibrate only where the failure is large and monotonic (short), not everywhere.

8 Limitations and Future Work

- **Ultra-short floor.** Even MoE is 86% at 25w with wide uncertainty. Training saw ≥50w; adding 10–35w + few-shot chat would likely beat thresholds.
- **Boundary blending.** 150w essays prefer baseline (97.4% vs 95.6%); soft routing (weighted P across short/long) could interpolate.
- **Threshold transfer.** Essay-tuned θ hurts human-heavy chat (–0.6 pp). Per-type thresholds (essay vs chat) are the next step.
- **Generators.** HC3 is one 2023 ChatGPT snapshot; modern holdout is 5 models. Paraphrase/adversarial eval is missing.
- **Scale.** Full 290k training might add 0.1–0.3 pp; transformers (DistilBERT/RoBERTa) with chunking remain the Phase-2 ceiling test.

Ethics: P(AI) is a screening signal, not proof. Short-text warnings and type reporting are required UI, not optional captions.

9 Conclusion

Length is not a detail — it is the task when evaluating short text. One global TF-IDF model buys 99% on essays by learning to call short text AI. Specializing by length, grounding short experts in real chat data, labeling chat/casual/essay, and calibrating per length recovers +10 to +50 pp where it matters, with a 0.3 pp full-text tax. We release the full stack so the claim is checkable: `moe/download_small_text.py` → `train_moe.py` → `calibrate_thresholds.py` →

`predict_moe.py`.

References

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- [8] art nitolog/llm-generated-texts (GPT-4 Turbo/Omni, Claude 3 Opus, Llama3, Command R+) — Hugging Face.
- [9] Mitchell, E. DetectGPT. arXiv:2301.11305 (2023) — zero-shot baseline family.

Appendices

A Hyperparameters

Component	Setting (all seeded)
Baseline word TF-IDF	(1,2) 25k, min_df 2, sublinear, clean_text_for_tfidf
Baseline char TF-IDF	char_wb (3,5) 50k, min_df 3, sublinear
Baseline classifiers	NB α=0.1; LR C=1.5 lbfgs 1000; LinearSVC C=1.0 + CalibratedCV(3)
Experts (each)	Word 15k + Char 25k $\rightarrow$ LR C=1.5 lbfgs 1000
Type gate	StandardScaler + LR C=1.0 lbfgs 1000 on 16 features

Table A1: Hyperparameters.

B Reproducibility checklist

- Datasplits: train subsample (10k/class, seed 42), 80/20 per pool (seed 42), calib 2.5k/eval 2.5k (seed 999), HC3/modern holdouts 15% (seed 42).
- Artifacts: `models/*.pkl` (baseline), `models/moe/expert_*.pkl`, `type_gate.pkl`, `thresholds.json`; metrics `results/moe_*.csv`.
- Indirection: truncation code is in `moe/type_features.py` (truncate_words/random_window); routing in `moe/train_moe.py::route_predict`.
- PDFs: `make_report.py` (6p), `moe/make_moe_appendix.py` (6p), `make_research_paper.py` (7p), this file (deep).

C Full routed table

Condition	Acc	Prec	Rec	F1	AUC
full	0.9896	0.9929	0.9790	0.9859	0.9990
first_25_words	0.8606	0.9259	0.6792	0.7836	0.9473
first_50_words	0.9312	0.9316	0.8794	0.9048	0.9796
first_100_words	0.9520	0.9140	0.9612	0.9370	0.9916
first_150_words	0.9558	0.9103	0.9774	0.9426	0.9946
first_200_words	0.9762	0.9662	0.9699	0.9680	0.9969
first_1_sent	0.8164	0.9066	0.5640	0.6954	0.9138
first_2_sent	0.9056	0.9288	0.8079	0.8641	0.9635
first_3_sent	0.9336	0.9253	0.8934	0.9091	0.9798
hc3_holdout	0.9852	0.9622	0.9906	0.9762	0.9990
modern_holdout	0.9570	0.9832	0.9649	0.9740	0.9891

Table C1: Routed MoE (results/moe_routed_metrics.csv).

Baseline counterpart in `results/moe_baseline_comparison.csv`; calibration detail in `results/moe_calibrated_metrics.csv`.